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the brain to the blood; and maintains a constant pressure inside the cranial vessels.

### 10.4.2 Spinal cord

The spinal cord is a long, slender, cylindrical nervous tissue. It is protected by the vertebral column and surrounded by the three membranes as in the brain. The spinal cord that extends from the brain stem into the vertebral canal of the vertebral column up to the level of 1st or 2nd lumbar vertebra. So the nerve roots of the remaining nerves are greatly elongated to exit the vertebral column at their appropriate space. The thick bundle of elongated nerve roots within the lower vertebral canal is called the **cauda equina** (horse's tail) because of its appearance.

In the cross section of spinal cord (Figure 10.11), there are two indentations: the posterior median sulcus and the anterior median fissure. Although there might be slight variations, the cross section of spinal cord is generally the same throughout its length. In contrast to the brain, the grey matter in the spinal cord forms an inner butterfly shaped region surrounded by the outer white matter. The grey matter consists of neuronal cell bodies and their dendrites, interneurons and glial cells. White matter consists of bundles of nerve fibres. In the center of the grey matter there is a central canal which is filled with CSF. Each half of the grey matter is divided into **a dorsal horn, a ventral horn** and **a lateral horn.**

The dorsal horn contains cell bodies of interneurons on which afferent neurons terminate. The ventral horn contains cell bodies of the efferent motor neurons supplying the skeletal muscle. Autonomic nerve fibres, supplying cardiac and smooth muscles and exocrine glands, originate from the cell bodies found in the lateral horn. In the white matter, the bundles of nerve fibres form two types of tracts namely **ascending tracts** which carry sensory impulses to the brain and descending **tracts** which carry motor impulses from the brain to the spinal nerves at various levels of the spinal cord. The spinal cord shows two enlargements, one in the cervical region and another one in the lumbosacral region. The **cervical**

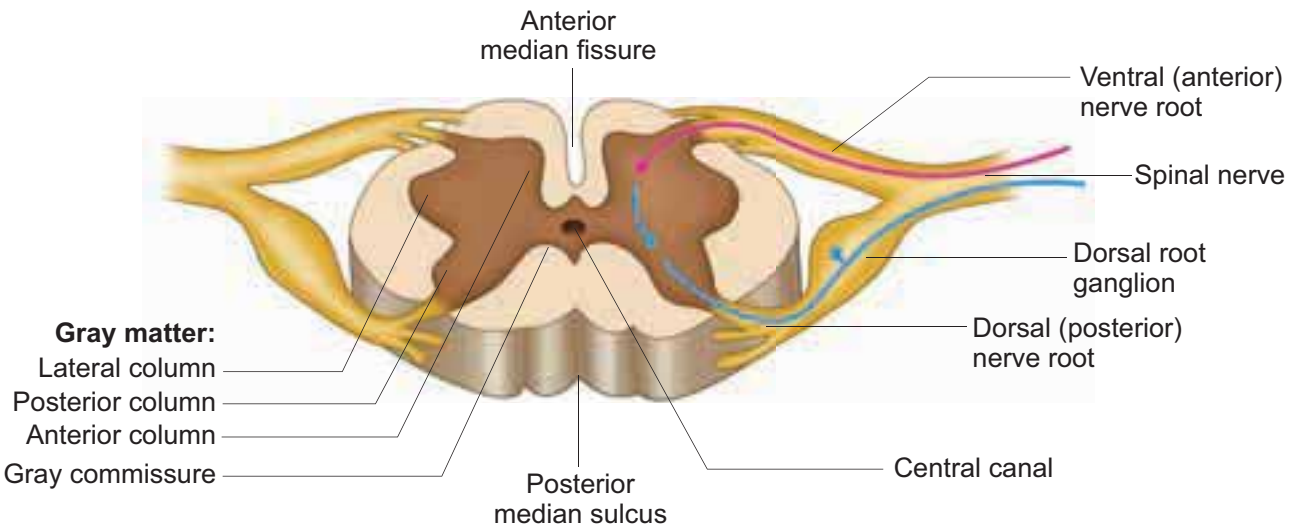


**Figure 10.11** C.S. of Spinal cord